

# Movie\_data\_analysis

November 5, 2025

```
[1]: import pandas as pd
import numpy as np
import seaborn as sns

import matplotlib.pyplot as plt
import matplotlib.mlab as mlab
import matplotlib
plt.style.use('ggplot')
from matplotlib.pyplot import figure

%matplotlib inline
matplotlib.rcParams['figure.figsize'] = (12,8)

pd.options.mode.chained_assignment = None

# Now we need to read in the data
df = pd.read_csv(r'movies.csv')
```

```
[2]: # Now let's take a look at the data

df.head()
```

```
[2]:
```

|   | name                                           | rating | genre     | year | \ |
|---|------------------------------------------------|--------|-----------|------|---|
| 0 | The Shining                                    | R      | Drama     | 1980 |   |
| 1 | The Blue Lagoon                                | R      | Adventure | 1980 |   |
| 2 | Star Wars: Episode V - The Empire Strikes Back | PG     | Action    | 1980 |   |
| 3 | Airplane!                                      | PG     | Comedy    | 1980 |   |
| 4 | Caddyshack                                     | R      | Comedy    | 1980 |   |

|   | released                      | score | votes     | director        | \ |
|---|-------------------------------|-------|-----------|-----------------|---|
| 0 | June 13, 1980 (United States) | 8.4   | 927000.0  | Stanley Kubrick |   |
| 1 | July 2, 1980 (United States)  | 5.8   | 65000.0   | Randal Kleiser  |   |
| 2 | June 20, 1980 (United States) | 8.7   | 1200000.0 | Irvin Kershner  |   |
| 3 | July 2, 1980 (United States)  | 7.7   | 221000.0  | Jim Abrahams    |   |
| 4 | July 25, 1980 (United States) | 7.3   | 108000.0  | Harold Ramis    |   |

|   | writer                  | star           | country        | budget \   |
|---|-------------------------|----------------|----------------|------------|
| 0 | Stephen King            | Jack Nicholson | United Kingdom | 19000000.0 |
| 1 | Henry De Vere Stacpoole | Brooke Shields | United States  | 4500000.0  |
| 2 | Leigh Brackett          | Mark Hamill    | United States  | 18000000.0 |
| 3 | Jim Abrahams            | Robert Hays    | United States  | 3500000.0  |
| 4 | Brian Doyle-Murray      | Chevy Chase    | United States  | 6000000.0  |

|   | gross       | company            | runtime |
|---|-------------|--------------------|---------|
| 0 | 46998772.0  | Warner Bros.       | 146.0   |
| 1 | 58853106.0  | Columbia Pictures  | 104.0   |
| 2 | 538375067.0 | Lucasfilm          | 124.0   |
| 3 | 83453539.0  | Paramount Pictures | 88.0    |
| 4 | 39846344.0  | Orion Pictures     | 98.0    |

```
[3]: # Let's see if there is any missing data

for col in df.columns:
    pct_missing = np.mean(df[col].isnull())
    print('{} - {}'.format(col, round(pct_missing*100)))
```

```
name - 0%
rating - 1%
genre - 0%
year - 0%
released - 0%
score - 0%
votes - 0%
director - 0%
writer - 0%
star - 0%
country - 0%
budget - 28%
gross - 2%
company - 0%
runtime - 0%
```

```
[4]: # Data Types for our columns

print(df.dtypes)
```

```
name          object
rating        object
genre         object
year          int64
released      object
score         float64
votes         float64
```

```

director    object
writer      object
star        object
country     object
budget      float64
gross       float64
company     object
runtime     float64
dtype: object

```

```

[5]: # change data type for column
df = df.dropna()

df['budget'] = df['budget'].astype('int64')

df['gross'] = df['gross'].astype('int64')

```

```

[6]: # Create correct year column

df['yearcorrect'] = df['released'].astype(str).str.extract(r'(\d{4})')

```

```

[7]: # Order our Data a little bit to see

df = df.sort_values(by=['gross'], inplace=False, ascending=False)

```

```

[8]: # Drop duplicates and clean budget/gross columns

df.drop_duplicates()

```

```

[8]:

```

|      | name                                       | rating    | genre  | year | \ |
|------|--------------------------------------------|-----------|--------|------|---|
| 5445 | Avatar                                     | PG-13     | Action | 2009 |   |
| 7445 | Avengers: Endgame                          | PG-13     | Action | 2019 |   |
| 3045 | Titanic                                    | PG-13     | Drama  | 1997 |   |
| 6663 | Star Wars: Episode VII - The Force Awakens | PG-13     | Action | 2015 |   |
| 7244 | Avengers: Infinity War                     | PG-13     | Action | 2018 |   |
| ...  | ...                                        | ...       | ...    | ...  |   |
| 5640 | Tanner Hall                                | R         | Drama  | 2009 |   |
| 2434 | Philadelphia Experiment II                 | PG-13     | Action | 1993 |   |
| 3681 | Ginger Snaps                               | Not Rated | Drama  | 2000 |   |
| 272  | Parasite                                   | R         | Horror | 1982 |   |
| 3203 | Trojan War                                 | PG-13     | Comedy | 1997 |   |

|      | released                          | score | votes     | \ |
|------|-----------------------------------|-------|-----------|---|
| 5445 | December 18, 2009 (United States) | 7.8   | 1100000.0 |   |
| 7445 | April 26, 2019 (United States)    | 8.4   | 903000.0  |   |
| 3045 | December 19, 1997 (United States) | 7.8   | 1100000.0 |   |
| 6663 | December 18, 2015 (United States) | 7.8   | 876000.0  |   |

|      |                                |     |          |
|------|--------------------------------|-----|----------|
| 7244 | April 27, 2018 (United States) | 8.4 | 897000.0 |
| ...  | ...                            | ... | ...      |
| 5640 | January 15, 2015 (Sweden)      | 5.8 | 3500.0   |
| 2434 | June 4, 1994 (South Korea)     | 4.5 | 1900.0   |
| 3681 | May 11, 2001 (Canada)          | 6.8 | 43000.0  |
| 272  | March 12, 1982 (United States) | 3.9 | 2300.0   |
| 3203 | October 1, 1997 (Brazil)       | 5.7 | 5800.0   |

|      | director            | writer                  | star \            |
|------|---------------------|-------------------------|-------------------|
| 5445 | James Cameron       | James Cameron           | Sam Worthington   |
| 7445 | Anthony Russo       | Christopher Markus      | Robert Downey Jr. |
| 3045 | James Cameron       | James Cameron           | Leonardo DiCaprio |
| 6663 | J.J. Abrams         | Lawrence Kasdan         | Daisy Ridley      |
| 7244 | Anthony Russo       | Christopher Markus      | Robert Downey Jr. |
| ...  | ...                 | ...                     | ...               |
| 5640 | Francesca Gregorini | Tatiana von Fürstenberg | Rooney Mara       |
| 2434 | Stephen Cornwell    | Wallace C. Bennett      | Brad Johnson      |
| 3681 | John Fawcett        | Karen Walton            | Emily Perkins     |
| 272  | Charles Band        | Alan J. Adler           | Robert Glaudini   |
| 3203 | George Huang        | Andy Burg               | Will Friedle      |

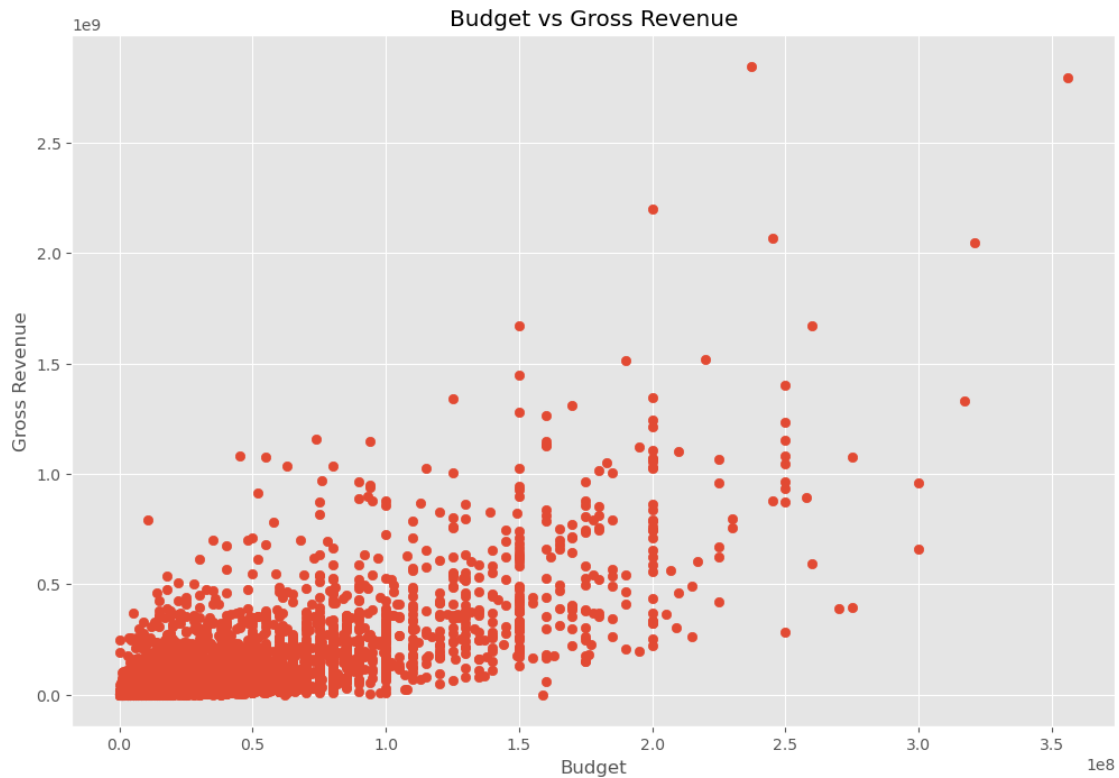
|      | country       | budget    | gross      | company \                 |
|------|---------------|-----------|------------|---------------------------|
| 5445 | United States | 237000000 | 2847246203 | Twentieth Century Fox     |
| 7445 | United States | 356000000 | 2797501328 | Marvel Studios            |
| 3045 | United States | 200000000 | 2201647264 | Twentieth Century Fox     |
| 6663 | United States | 245000000 | 2069521700 | Lucasfilm                 |
| 7244 | United States | 321000000 | 2048359754 | Marvel Studios            |
| ...  | ...           | ...       | ...        | ...                       |
| 5640 | United States | 3000000   | 5073       | Two Prong Lesson          |
| 2434 | United States | 5000000   | 2970       | Trimark Pictures          |
| 3681 | Canada        | 5000000   | 2554       | Copperheart Entertainment |
| 272  | United States | 800000    | 2270       | Embassy Pictures          |
| 3203 | United States | 15000000  | 309        | Daybreak                  |

|      | runtime | yearcorrect |
|------|---------|-------------|
| 5445 | 162.0   | 2009        |
| 7445 | 181.0   | 2019        |
| 3045 | 194.0   | 1997        |
| 6663 | 138.0   | 2015        |
| 7244 | 149.0   | 2018        |
| ...  | ...     | ...         |
| 5640 | 96.0    | 2015        |
| 2434 | 97.0    | 1994        |
| 3681 | 108.0   | 2001        |
| 272  | 85.0    | 1982        |
| 3203 | 85.0    | 1997        |

[5421 rows x 16 columns]

```
[9]: # Scatter plots with budget vs gross

plt.scatter(x=df['budget'], y=df['gross'])
plt.title('Budget vs Gross Revenue')
plt.xlabel('Budget')
plt.ylabel('Gross Revenue')
plt.show()
```



```
[10]: # Plot budget vs gross using seaborn
print(df.dtypes)
sns.regplot(x='budget', y='gross', data=df, scatter_kws={"color": "red"},
            line_kws={"color": "blue"})
```

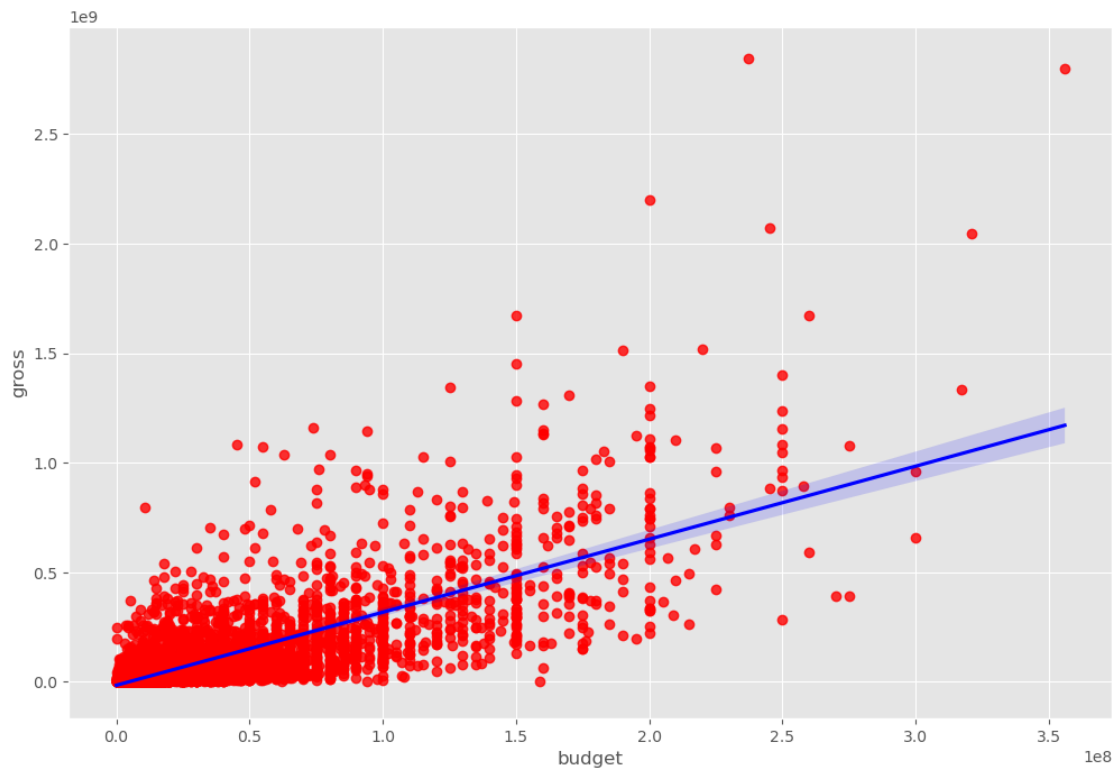
|          |         |
|----------|---------|
| name     | object  |
| rating   | object  |
| genre    | object  |
| year     | int64   |
| released | object  |
| score    | float64 |
| votes    | float64 |

```

director      object
writer        object
star          object
country       object
budget        int64
gross         int64
company       object
runtime       float64
yearcorrect   object
dtype: object

```

```
[10]: <Axes: xlabel='budget', ylabel='gross'>
```



```

[11]: # Correlations
numeric_df = df.select_dtypes(include=['float64', 'int64', 'Int64'])
corr_matrix = numeric_df.corr()
print(corr_matrix)

```

|        | year     | score    | votes    | budget   | gross    | runtime  |
|--------|----------|----------|----------|----------|----------|----------|
| year   | 1.000000 | 0.056386 | 0.206021 | 0.327722 | 0.274321 | 0.075077 |
| score  | 0.056386 | 1.000000 | 0.474256 | 0.072001 | 0.222556 | 0.414068 |
| votes  | 0.206021 | 0.474256 | 1.000000 | 0.439675 | 0.614751 | 0.352303 |
| budget | 0.327722 | 0.072001 | 0.439675 | 1.000000 | 0.740247 | 0.318695 |

```
gross    0.274321  0.222556  0.614751  0.740247  1.000000  0.275796
runtime  0.075077  0.414068  0.352303  0.318695  0.275796  1.000000
```

```
[12]: # High correlation between gross and budget

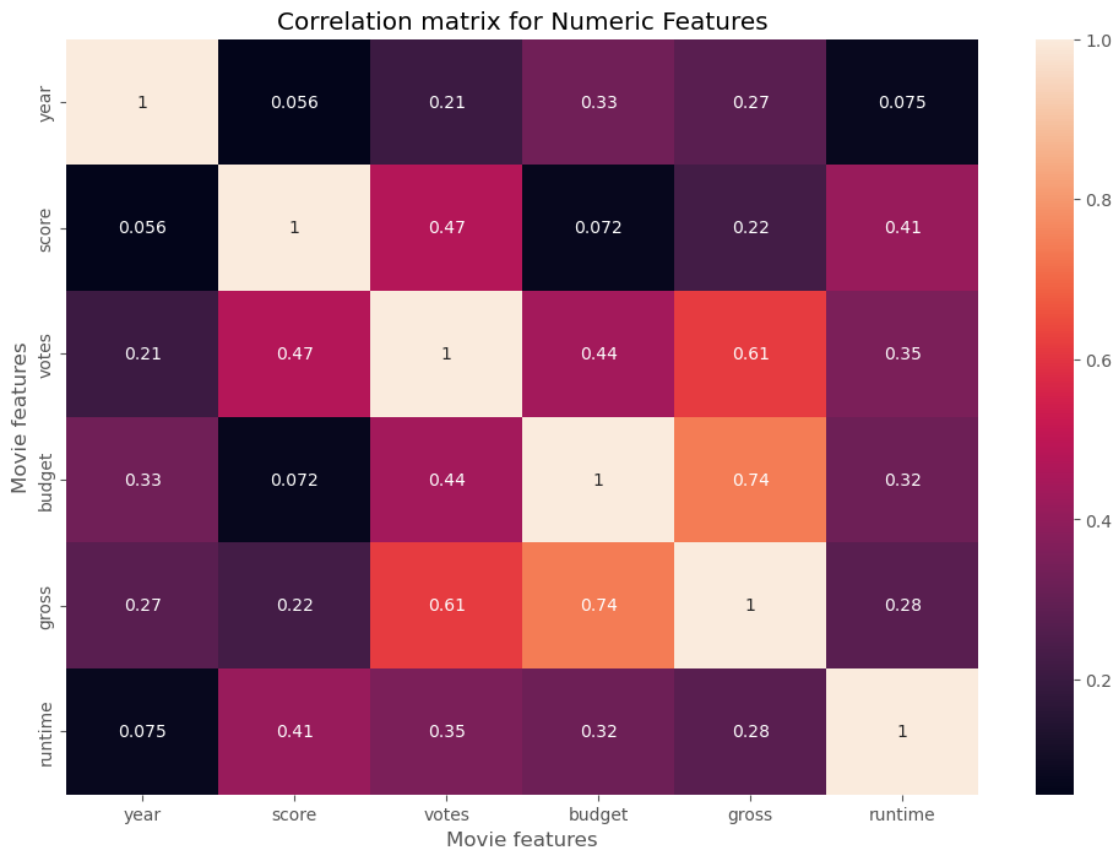
sns.heatmap(corr_matrix, annot = True)

plt.title("Correlation matrix for Numeric Features")

plt.xlabel("Movie features")

plt.ylabel("Movie features")

plt.show()
```



```
[13]: df_numerized = df

for col in df_numerized.columns:
    if df_numerized[col].dtype == 'object':
        df_numerized[col] = df_numerized[col].astype('category')
```

```
df_numerized[col] = df_numerized[col].astype('category').cat.codes
```

```
df_numerized
```

```
[13]:
```

|      | name | rating | genre | year | released | score | votes     | director | writer | \   |
|------|------|--------|-------|------|----------|-------|-----------|----------|--------|-----|
| 5445 | 386  | 5      | 0     | 2009 | 527      | 7.8   | 1100000.0 | 785      | 1263   |     |
| 7445 | 388  | 5      | 0     | 2019 | 137      | 8.4   | 903000.0  | 105      | 513    |     |
| 3045 | 4909 | 5      | 6     | 1997 | 534      | 7.8   | 1100000.0 | 785      | 1263   |     |
| 6663 | 3643 | 5      | 0     | 2015 | 529      | 7.8   | 876000.0  | 768      | 1806   |     |
| 7244 | 389  | 5      | 0     | 2018 | 145      | 8.4   | 897000.0  | 105      | 513    |     |
| ...  | ...  | ...    | ...   | ...  | ...      | ...   | ...       | ...      | ...    | ... |
| 5640 | 3794 | 6      | 6     | 2009 | 890      | 5.8   | 3500.0    | 585      | 2924   |     |
| 2434 | 2969 | 5      | 0     | 1993 | 1467     | 4.5   | 1900.0    | 1805     | 3102   |     |
| 3681 | 1595 | 3      | 6     | 2000 | 1721     | 6.8   | 43000.0   | 952      | 1683   |     |
| 272  | 2909 | 6      | 9     | 1982 | 1525     | 3.9   | 2300.0    | 261      | 55     |     |
| 3203 | 4966 | 5      | 4     | 1997 | 2152     | 5.7   | 5800.0    | 651      | 161    |     |

|      | star | country | budget    | gross      | company | runtime | yearcorrect |
|------|------|---------|-----------|------------|---------|---------|-------------|
| 5445 | 1534 | 47      | 237000000 | 2847246203 | 1382    | 162.0   | 29          |
| 7445 | 1470 | 47      | 356000000 | 2797501328 | 983     | 181.0   | 39          |
| 3045 | 1073 | 47      | 200000000 | 2201647264 | 1382    | 194.0   | 17          |
| 6663 | 356  | 47      | 245000000 | 2069521700 | 945     | 138.0   | 35          |
| 7244 | 1470 | 47      | 321000000 | 2048359754 | 983     | 149.0   | 38          |
| ...  | ...  | ...     | ...       | ...        | ...     | ...     | ...         |
| 5640 | 1498 | 47      | 3000000   | 5073       | 1385    | 96.0    | 35          |
| 2434 | 186  | 47      | 5000000   | 2970       | 1376    | 97.0    | 14          |
| 3681 | 527  | 6       | 5000000   | 2554       | 466     | 108.0   | 21          |
| 272  | 1473 | 47      | 800000    | 2270       | 582     | 85.0    | 2           |
| 3203 | 1811 | 47      | 15000000  | 309        | 504     | 85.0    | 17          |

```
[5421 rows x 16 columns]
```

```
[14]: # Correlations
print(df_numerized.corr())

sns.heatmap(df_numerized.corr(), annot = True)

plt.title("Correlation matrix")

plt.xlabel("Movie features")

plt.ylabel("Movie features")

plt.show()
```

|      | name     | rating    | genre    | year     | released  | score    | \ |
|------|----------|-----------|----------|----------|-----------|----------|---|
| name | 1.000000 | -0.029234 | 0.010996 | 0.025542 | -0.006152 | 0.014450 |   |

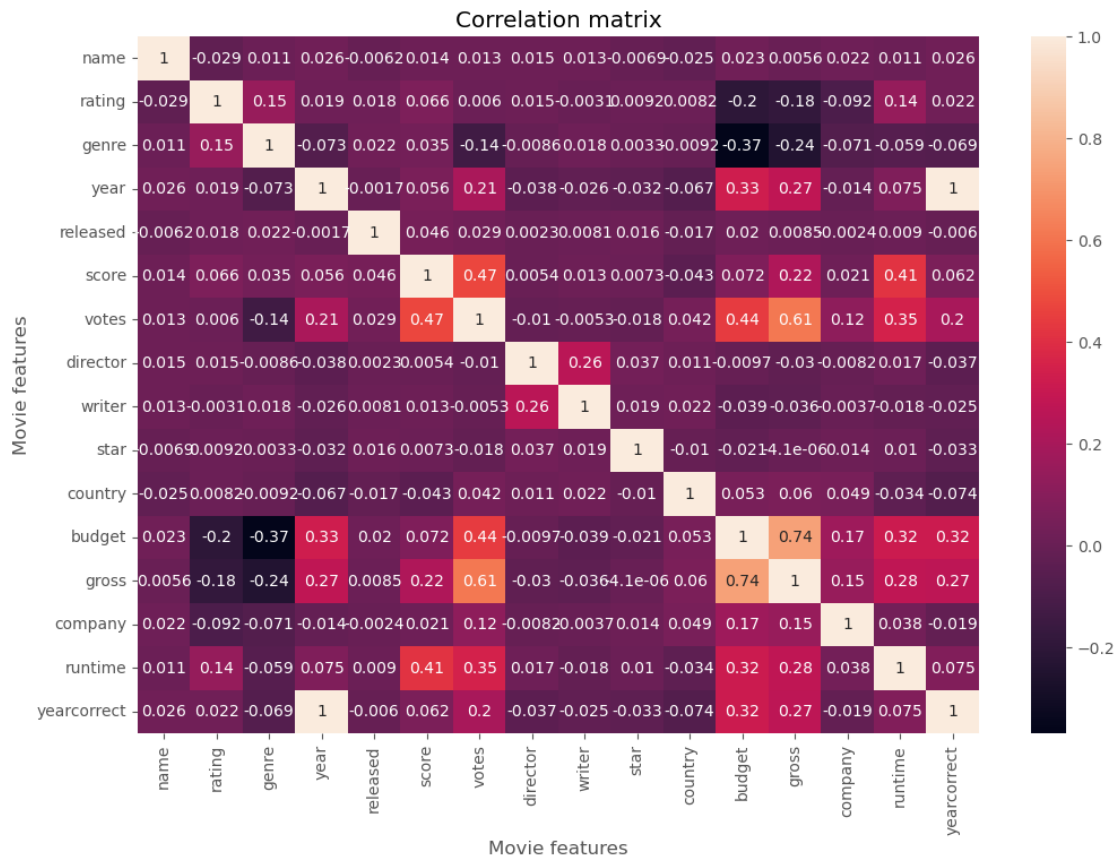


|             |           |           |           |           |           |           |
|-------------|-----------|-----------|-----------|-----------|-----------|-----------|
| rating      | -0.029234 | 1.000000  | 0.147796  | 0.019499  | 0.018083  | 0.065983  |
| genre       | 0.010996  | 0.147796  | 1.000000  | -0.073167 | 0.022142  | 0.035106  |
| year        | 0.025542  | 0.019499  | -0.073167 | 1.000000  | -0.001740 | 0.056386  |
| released    | -0.006152 | 0.018083  | 0.022142  | -0.001740 | 1.000000  | 0.045874  |
| score       | 0.014450  | 0.065983  | 0.035106  | 0.056386  | 0.045874  | 1.000000  |
| votes       | 0.012615  | 0.006031  | -0.135990 | 0.206021  | 0.028833  | 0.474256  |
| director    | 0.015246  | 0.014656  | -0.008553 | -0.038354 | 0.002308  | 0.005413  |
| writer      | 0.012880  | -0.003149 | 0.017578  | -0.025908 | 0.008072  | 0.012843  |
| star        | -0.006882 | 0.009196  | 0.003341  | -0.032157 | 0.015706  | 0.007296  |
| country     | -0.025490 | 0.008230  | -0.009164 | -0.066748 | -0.017228 | -0.043051 |
| budget      | 0.023392  | -0.203946 | -0.368523 | 0.327722  | 0.019952  | 0.072001  |
| gross       | 0.005639  | -0.181906 | -0.244101 | 0.274321  | 0.008501  | 0.222556  |
| company     | 0.021697  | -0.092357 | -0.071334 | -0.014333 | -0.002407 | 0.020656  |
| runtime     | 0.010850  | 0.140792  | -0.059237 | 0.075077  | 0.008975  | 0.414068  |
| yearcorrect | 0.025542  | 0.022021  | -0.069147 | 0.998726  | -0.005989 | 0.061923  |

|             |           |           |           |           |           |           |   |
|-------------|-----------|-----------|-----------|-----------|-----------|-----------|---|
|             | votes     | director  | writer    | star      | country   | budget    | \ |
| name        | 0.012615  | 0.015246  | 0.012880  | -0.006882 | -0.025490 | 0.023392  |   |
| rating      | 0.006031  | 0.014656  | -0.003149 | 0.009196  | 0.008230  | -0.203946 |   |
| genre       | -0.135990 | -0.008553 | 0.017578  | 0.003341  | -0.009164 | -0.368523 |   |
| year        | 0.206021  | -0.038354 | -0.025908 | -0.032157 | -0.066748 | 0.327722  |   |
| released    | 0.028833  | 0.002308  | 0.008072  | 0.015706  | -0.017228 | 0.019952  |   |
| score       | 0.474256  | 0.005413  | 0.012843  | 0.007296  | -0.043051 | 0.072001  |   |
| votes       | 1.000000  | -0.010376 | -0.005316 | -0.017638 | 0.041551  | 0.439675  |   |
| director    | -0.010376 | 1.000000  | 0.261735  | 0.036593  | 0.011133  | -0.009662 |   |
| writer      | -0.005316 | 0.261735  | 1.000000  | 0.018520  | 0.022488  | -0.039466 |   |
| star        | -0.017638 | 0.036593  | 0.018520  | 1.000000  | -0.009990 | -0.021473 |   |
| country     | 0.041551  | 0.011133  | 0.022488  | -0.009990 | 1.000000  | 0.052977  |   |
| budget      | 0.439675  | -0.009662 | -0.039466 | -0.021473 | 0.052977  | 1.000000  |   |
| gross       | 0.614751  | -0.029560 | -0.035885 | -0.000004 | 0.060078  | 0.740247  |   |
| company     | 0.118470  | -0.008223 | -0.003697 | 0.014082  | 0.048569  | 0.170235  |   |
| runtime     | 0.352303  | 0.017433  | -0.017561 | 0.010108  | -0.034477 | 0.318695  |   |
| yearcorrect | 0.203098  | -0.037371 | -0.025495 | -0.032687 | -0.073569 | 0.320312  |   |

|          |           |           |           |             |
|----------|-----------|-----------|-----------|-------------|
|          | gross     | company   | runtime   | yearcorrect |
| name     | 0.005639  | 0.021697  | 0.010850  | 0.025542    |
| rating   | -0.181906 | -0.092357 | 0.140792  | 0.022021    |
| genre    | -0.244101 | -0.071334 | -0.059237 | -0.069147   |
| year     | 0.274321  | -0.014333 | 0.075077  | 0.998726    |
| released | 0.008501  | -0.002407 | 0.008975  | -0.005989   |
| score    | 0.222556  | 0.020656  | 0.414068  | 0.061923    |
| votes    | 0.614751  | 0.118470  | 0.352303  | 0.203098    |
| director | -0.029560 | -0.008223 | 0.017433  | -0.037371   |
| writer   | -0.035885 | -0.003697 | -0.017561 | -0.025495   |
| star     | -0.000004 | 0.014082  | 0.010108  | -0.032687   |
| country  | 0.060078  | 0.048569  | -0.034477 | -0.073569   |
| budget   | 0.740247  | 0.170235  | 0.318695  | 0.320312    |
| gross    | 1.000000  | 0.149553  | 0.275796  | 0.268721    |

|             |          |           |          |           |
|-------------|----------|-----------|----------|-----------|
| company     | 0.149553 | 1.000000  | 0.037585 | -0.018806 |
| runtime     | 0.275796 | 0.037585  | 1.000000 | 0.075294  |
| yearcorrect | 0.268721 | -0.018806 | 0.075294 | 1.000000  |



```
[15]: mat_corr = df_numerized.corr()

corr_pairs = mat_corr.unstack()

print(corr_pairs)
```

|             |             |           |
|-------------|-------------|-----------|
| name        | name        | 1.000000  |
|             | rating      | -0.029234 |
|             | genre       | 0.010996  |
|             | year        | 0.025542  |
|             | released    | -0.006152 |
|             | ...         |           |
| yearcorrect | budget      | 0.320312  |
|             | gross       | 0.268721  |
|             | company     | -0.018806 |
|             | runtime     | 0.075294  |
|             | yearcorrect | 1.000000  |

Length: 256, dtype: float64

```
[16]: sorted_pairs = corr_pairs.sort_values()
      print(sorted_pairs)
```

```
genre      budget      -0.368523
budget     genre      -0.368523
gross      genre      -0.244101
genre      gross      -0.244101
rating     budget     -0.203946
...
year       year       1.000000
genre      genre      1.000000
rating     rating     1.000000
runtime    runtime    1.000000
yearcorrect yearcorrect 1.000000
Length: 256, dtype: float64
```

```
[17]: # high correlation

high_corr = sorted_pairs[(sorted_pairs > 0.5)]
print(high_corr)
```

```
gross      votes      0.614751
votes      gross      0.614751
gross      budget     0.740247
budget     gross      0.740247
year       yearcorrect 0.998726
yearcorrect year      0.998726
name       name       1.000000
company    company    1.000000
gross      gross      1.000000
budget     budget     1.000000
country    country    1.000000
star       star       1.000000
writer     writer     1.000000
director   director    1.000000
votes      votes      1.000000
score      score      1.000000
released   released    1.000000
year       year       1.000000
genre      genre      1.000000
rating     rating     1.000000
runtime    runtime    1.000000
yearcorrect yearcorrect 1.000000
dtype: float64
```

```
[18]: # votes and budgets have the highest correlation
```